

The Customer Twin: System Overview

DATA INPUTS

MRI Event Logs

40 scanners · Apr–May 2024

Patient Demographics

Age · Weight · Height · Direction

Coil State Signals

MRI_CCS_11 · BC · SP1–SP8 · HE1–HE4

Hardware Signals

Tube life · Arcing events · FRR codes

Body Region Sequences

HEAD · SPINE · PELVIS · 11 regions

CUSTOMER TWIN

Unified Transformer Architecture

Orchestration Model

Day-level body region sequencing

Exchange Model

Patient setup & transitions

Examination Model

MRI scan sequence generation

$\mu \pm \sigma$ Uncertainty Quantification

Per-token duration prediction

OUTPUTS

Synthetic Schedules

Full simulated days with timestamps

ROI Simulation

Model coil/upgrade impact pre-purchase

Predictive Maintenance

2-week advance failure warning

Semantic Insight Agent

R&D pain extraction from comments

VR Training Simulation

Unity · Meta Quest technician training

White-box AI: all predictions include uncertainty bounds and interpretable conditioning features